

The ScholarMaster Architecture: A Unified Reference Model for Privacy-First Intelligent Campus Systems

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Abstract—Current research in intelligent cyber-physical infrastructure frequently suffers from a fragmentation challenge: hardware efficiency, algorithmic accuracy, data privacy, and sociological trust are typically treated as isolated domains. This paper introduces Constraint-First Architectural Synthesis (CFAS) to define a unified reference architecture for privacy-centered edge intelligence. We construct a complete vertically layered stack—from thermal equilibrium in edge silicon to sociotechnical legitimacy constraints—and formalize the Canonical Invariant Namespace (INV-01 to INV-15) that binds these layers together. By mapping the operational adversary model (A_0–A_4) and tracing the end-to-end lifecycle of a spatiotemporal event across the eight-layer boundary using formal sequence diagrams and schema-constrained semantic firewalls, this framework demonstrates how irreversible abstraction, computationally interposed governance, and visible privacy constraints converge. This paper establishes a formal doctrine for cross-layer operational integration, proving that privacy and deployment feasibility depend inherently on architecturally bounded computation.

Keywords—Constraint-First Architectural Synthesis, Cross-Layer Execution Contracts, Irreversible Abstraction Boundaries, Governance-Aware Deployment, Cyber-Physical Integration, Semantic Firewalls.

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Current research in intelligent cyber-physical infrastructure frequently suffers from a fragmentation challenge: hardware efficiency, algorithmic accuracy, data privacy, and sociological trust are typically treated as isolated domains. This paper introduces Constraint-First Architectural Synthesis (CFAS) to define a unified reference architecture for privacy-centered edge intelligence. We construct a complete vertically layered stack—from thermal equilibrium in edge silicon to sociotechnical legitimacy constraints—and formalize the Canonical Invariant Namespace (INV-01 to INV-15) that binds these layers together. By mapping the operational adversary model (A_0–A_4) and tracing the end-to-end lifecycle of a spatiotemporal event across the eight-layer boundary using formal sequence diagrams and schema-constrained semantic firewalls, this framework demonstrates how irreversible abstraction, computationally interposed governance, and visible privacy constraints converge. This paper establishes a formal doctrine for cross-layer operational integration, proving that privacy and deployment feasibility depend inherently on architecturally bounded computation.

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INTRODUCTION

The development of intelligent academic environments often encounters a significant integration barrier. The dominant architectural thesis of this work is Constraint-First Architectural Synthesis (CFAS). We argue that operational trust emerges from layered constraint enforcement, not isolated subsystem optimization. Within the ScholarMaster architecture, this subsystem synthesizes architectural layers and cross-layer execution logic. It does not derive formal verification theorems (Paper 21), implement runtime watchdog semantics (Paper 18), optimize federated convergence (Paper 14), or conduct sociological longitudinal analysis (Paper 16).

Limitations of Isolated Optimization Paradigms

In traditional deployments, system properties are optimized in isolation. Computer vision pipelines maximize accuracy while ignoring the severe thermal constraints of edge deployment; stream processing engines optimize throughput while neglecting the sociotechnical requirements of user trust; and cryptographic privacy protocols propose abstractions that are computationally infeasible on constrained embedded hardware. Isolated optimization leads to operational incoherence. Current architectures are characterized by an overreliance on centralized cloud-based trust, a lack of explicit privacy-preserving execution contracts, the separation of governance and computation, and insufficient reconciliation of physical limitations with sociotechnical legitimacy. Most critically, privacy is inherently compromised when these approaches fail to enforce a unified architectural framework.

Constraint-First Architecture

The ScholarMaster Architecture deals with the problem of fragmentation by reconciling the constraints at various layers. It is not a general-purpose modular AI design, but rather a constrained vertical solution, which binds the real-time requirements, privacy regulations, physical limitations of its supporting hardware, and the social costs of deployment into a unified whole. It is important to note that the constraints are prioritized during the reconciliation process. The architecture defines the invariants that must be satisfied by any implementation that uses its constrained execution model and

establishes a correspondence between theoretical results and their practical manifestations.

CONSTRAINT-FIRST ARCHITECTURAL SYNTHESIS (CFAS)

To characterize the architectural constraints, CFAS defines a unified admissibility space. Let $\mathcal{C}$ denotes the set of admissible computational states. A computation is admissible iff it satisfies a conjunction of cross-layer constraints:

Equation: $\mathcal{C}_{\text{admissible}} = C_{\text{phys}} \wedge C_{\text{therm}} \wedge C_{\text{priv}} \wedge C_{\text{gov}} \wedge C_{\text{time}}$

Where C_{phys} bounds execution to local hardware topologies, C_{therm} prevents silicon degradation, C_{priv} enforces irreversible dimensional reduction, C_{gov} requires cryptographic authorization For egress, and C_{time} bounds temporal exposure in volatile memory. Under CFAS, any computation which violates any constraint is deemed architecturally invalid, and is not tolerated by the system - it fails gracefully rather than being allowed to proceed to an erroneous state. The trick to CFAS is that these limits are combined together in a single edge-stack.

THE THREAT LANDSCAPE AND ADVERSARY MODEL

To put forward the architectural requirements formulated in later sections, we use the formal adversary model from Paper 19. The ScholarMaster architecture is organized in such a way that each level corresponds to a particular type of adversary, which is formalized there.

Adversary Classes and Attack Vectors

- **A_0 (Passive Observer):** An entity observing the physical environment. Defended via: Transparent signaling paradigms (Paper 15).
- **A_1 (Network Eavesdropper):** An entity actively sniffing packet traffic. Defended via: Mutual TLS (mTLS) and Differential Privacy (Papers 11, 13).
- **A_2 (Malicious/Curious Administrator):** An insider attempting retrospective mass surveillance. Defended via: The absolute L3 irreversibility boundary which destroys raw data before persistence (Papers 3, 8).
- **A_3 (Hardware Thief / Root Access):** An attacker attempting physical OS-level extraction. Defended via: Enforced architectural volatility and fail-closed hardware supervision bounds (Papers 11, 12, 18).
- **A_4 (State-Level Hardware Attacker):** Explicitly out of scope for the ScholarMaster architecture as formalized in Paper 19.

The architectural invariants presented in this reference model are designed strictly to neutralize adversaries up to and including class A_3.

THE SCHOLARMASTER REFERENCE STACK

The architecture is organized into four distinct strata. While Layers 1-8 represent the data flow, the Research Series papers map to these strata to provide operational, logical, and sociological definitions.

[FIGURE: Architectural Pipeline / Validation Curves]

$\caption{The ScholarMaster Unified Reference Stack. The system is built upwards from physical constraints (I) to logical reasoning (II), verified trust (III), and finally social acceptance (IV).}$

$\label{fig:stack}$
 $\end{figure}$

Stratum I: The Physics of Intelligence & Operations

This layer concerns the physical, thermal, and operational reality of the edge node. It enforces "Privacy by Physics" and operational

resilience against harsh deployment conditions.

- **L1 Physical Substrate:** Determined by Paper 11 (Production MLOps) and Paper 12 (Flash Endurance). It defines the operational time of the runtime: enforces memory volatility, execution supervision, and filesystem immutability.
- **L2 Sensor Acquisition:** Addressed in Paper 5 (Thermal/Memory) and Paper 10 (System level validation). Included in this category is Paper 6 (Acoustic anomaly) which describes high fidelity environmental capture confined to a circular volatile buffer.

Stratum II: The Logic of Stewardship

This layer converts raw signals into semantic meaning and governance compliance without reconstructing identity.

- **L3 Edge Abstraction:** Defined by Paper 17 (Irreversibility Doctrine) and Paper 3 (Pose Abstraction). This is the absolute "Irreversibility Boundary." High-entropy raw matrices are destructively compressed into low-dimensional metadata vectors.
- **L4 Inference:** Validated in Paper 1 (Biometric HNSW), Paper 2 (Context), and Paper 4 (Schedule Compliance). Execution here is identity-blind, context-aware, and strictly bound by the spatiotemporal continuous verification engine (Paper 7).

Stratum III: The Architecture of Verification

This layer provides the proofs that the system is behaving as claimed, bridging local logic with global accountability.

- **L5 Governance:** Paper 9 (Orchestration). Enforces policy gates before any data emission. It physically separates the act of algorithmic judgment (L4) from the authorization to transmit.
- **L6 Human Output:** Paper 15 (AR Visualization) and Paper 16 (Sociological Trust). Translates machine boundaries into human-verifiable proof via skeleton-only augmented reality displays.
- **L7 Audit Persistence:** Paper 8 (Blockchain Provenance). Implements a tamper-evident, append-only local ledger ensuring that A_2 adversaries cannot retroactively alter compliance histories.

Stratum IV: The Sociology of Trust and Federation

This layer treats sociotechnical legitimacy as an operational systems constraint, not an ethical afterthought. Institutional acceptance materially affects deployability; visible boundedness improves operational trust.

- L8 Federation: Allows the network to learn global behavioral drifts without transmitting local data, governed by strict differential privacy budgets. This layer preserves federated institutional decentralization.

THE CANONICAL INVARIANT NAMESPACE

To ensure consistency across the 19-paper research program, we formalize the ScholarMaster Invariant Namespace.

Contrastingly, the invariant namespace does not enforce any new constraints. Instead, it collects all the checked constraints from the component studies (P1, P3, P9, P17, P18). These invariants are formal contracts between the system layers: a violation of any invariant results in an immediate L1 hardware exception.

Ephemeral Data Constraints

- INV-01 (Raw Non-Persistence): No sensor data shall persist beyond the L3 boundary (Volatile RAM) [Ref].
- INV-04 (Bounded Temporal Exposure): Any buffer containing raw data must be explicitly zeroed within a strict temporal boundary (detailed in Paper 18).
- INV-09 (Volatile-Only Processing): All L1-L4 execution must occur exclusively in volatile memory, structurally preventing state persistence to disk or swap space [Ref].
- INV-10 (Density Constraint): L3 output dimensionality is strictly capped to ensure abstract irreversibility, formally preventing the inversion of the neural network back to pixel space [Ref].
- INV-15 (Privacy Mode Default): Following any reboot or power loss, the system inevitably boots into restricted Privacy Mode, requiring an explicit, cryptographically signed remote handshake to elevate privileges [Ref].

Governance & Control Flow

- INV-02 (Identity Non-Propagation): Identity tokens are ephemeral, randomly generated per session, and valid only for the immediate execution context [Ref].
- INV-03 (Governance Non-Bypass): No data, even abstract metadata, can transition to the network egress module without passing L5 Governance logic checks [Ref].
- INV-07 (Audit Immutability): All governance decisions (accept/reject) must be cryptographically hashed and logged to the immutable ledger before the corresponding physical or network action is executed [Ref].
- INV-11 (Consent-Gated Enrollment): No biometric or behavioral embedding is generated or stored without verified, non-expired, and digitally signed user consent [Ref].

Distributed Sovereignty

- INV-05 (Federation Sovereignty): No cross-campus model weight update shall contain gradient data allowing the mathematical reconstruction of localized training samples [Ref].
- INV-12 (Deletion Propagation): User data deletion/revocation requests must propagate to the global federation state, forcing an unlearning routine within one aggregation round [Ref].
- INV-13 (Federation Payload Restriction): L8 egress transmits only DP-noised gradient summaries; explicit data types are banned at the gRPC schema level [Ref].

Operational Resilience

- INV-06 (Thermal Equilibrium): Inference loads must be dynamically throttled to maintain component temperatures strictly below physical silicon degradation limits [Ref].
- INV-08 (Fail-Closed Liveness): System and execution failures inevitably transition the node to a HALT/Reboot state, never an OPEN/permissive bypass state [Ref].
- INV-14 (Non-Disableable Enforcement): Lifecycle constraints and bounds are structurally enforced by the runtime architecture and

INTER-STRATUM EXECUTION CONTRACTS

To meet architectural requirements, the system uses schema-constrained semantic firewalls. In traditional applications, the architecture passes either opaque pointers or dynamically typed JSON objects between microservices, thereby facilitating unconstrained data propagation. The architecture below utilizes type-rich Protocol Buffer (ProtoBuf) schemas to enforce information discipline at the system's layer. Notably, schemas are used as implementation details rather than intermediate representations. The approach allows the system to maintain architectural isolation because Protobuf schemas dictate what data structures each layer is allowed to pass. Moreover, it becomes physically impossible to pass fields that were not anticipated in the destination structure.

Semantic Firewalls at the Execution Boundary

Communication between Stratum I (Acquisition) and Stratum II (Logic) occurs across a strict semantic firewall. The execution contract ensures that reasoning logic evaluates the abstraction without ever assuming ownership of the raw data memory space. Irreversible representation boundaries are compiled into the payload definitions.

Listing 1 defines the Canonical Protobuf Schema utilized to bridge the Irreversibility Boundary. By definition, the schema lacks any fields capable of holding pixel bytes or string-based personal identifiers.

[FIGURE: Architectural Pipeline / Validation Curves]

The Temporal Budget Contract

Rather than simply defining data structures, the architecture further imposes a temporally-globally verifiable execution contract on the processing pipeline; the runtime engine must finish processing from the sensor capture (L2) through abstraction (L3), logical evaluation (L4) and buffer destruction within the temporal constraint ($\Delta t \leq 33\text{ms}$) of the contract. Should the execution engine fail to perform an atomic commit due to processing saturation or erroneous computation, the stratum I supervisory function will perform a forced kill of the volatile execution context, thereby preventing potential information disclosure, and recording the execution fault on the

ledger.

THE LIFE OF AN EVENT: END-TO-END SEQUENCE

To illustrate the integration of the 19 sub-systems, we map the lifecycle of a spatiotemporal event across the architecture using a formal sequence diagram (Figure \ref{fig:sequence}).

[FIGURE: Architectural Pipeline / Validation Curves]



Consider a student entering a restricted examination room:

- Ingestion (L1/L2): Photons hit the camera sensor. A frame is ingested into a bounded volatile memory buffer. The operational runtime initiates the strict temporal limit (INV-04).
- Abstraction (L3): The frame is processed by the NPU. A lightweight CNN extracts abstract geometric keypoints (INV-10). Crucial step: The original frame in the volatile buffer is physically overwritten with zeros (INV-01).
- Reasoning (L4): The abstract `AbstractEvent` protobuf is evaluated by the continuous stream processing engine. The engine checks spatiotemporal invariants and queries the local schedule.
- Governance (L5): The positive compliance match is packaged into an abstract metadata payload. The Governance Filter verifies that active institutional policy permits logging attendance at this specific time (INV-03).
- Audit (L7): The governance decision is hashed using SHA-256 and appended to the local tamper-evident blockchain ledger (INV-07), securing the decision against insider tampering.
- Hardware: Lightweight IP cameras, streaming RSTP to local server rack (Macro-Edge) located at the building’s utility closet. In the case of hardware replacement asymmetry, the computational power should be placed at the center (not in multiple camera nodes).
- Layer Mapping: L1/L2 reside on the legacy IP cameras. L3 through L7 execute on a high-throughput UMA compute cluster. Operational maintenance overhead is centralized.
- Security Boundary: The RTSP stream is on a non-routable VLAN. The Macro-Edge server is the sole L3 semantic firewall, actively dropping packets that threaten to exceed thermal or temporal processing deadlines, thus keeping the raw footage securely contained within the local closet.

THE THEOREM-IMPLEMENTATION LATTICE

The formal algebra and theorems are fully defined in the foundational papers (P17-P19). This document contains no new formal assertions or proofs. Table I provides a cross-section of the fundamental principles corresponding to the system engineering implementations validated in Papers 1-18.

TABLE: Mapping Formal Principles to System Implementation

PHYSICAL DEPLOYMENT TOPOLOGIES

A reference architecture is incomplete without specifying its physical instantiation mapping. The eight-layer logical stack maps fluidly to two primary institutional deployment profiles:

Profile A: The Autonomous Micro-Edge

Designed for bandwidth-constrained campuses facing intermittent power stability.

- Hardware: A single integrated System-on-Chip (e.g., Jetson Nano) mounted inside the classroom. Procurement limitations frequently necessitate aging hardware, imposing severe thermal constraints.
- Layer Mapping: Layers L1 through L7 execute entirely on the single physical node. The local SQLite buffer maintains the L7 ledger under constrained I/O capacity.
- Network Dependency: Zero. The system operates autonomously, enforcing all governance rules locally and syncing to L8 only when external networks are restored, masking physical network instability from operational logic.

Profile B: The Campus Macro-Edge Cluster

Designed for high-density environments navigating complex campus-network segmentation.

LIMITATIONS

The constraint-first synthesis of this reference architecture inherently bounds its deployment feasibility and evolution:

- Deployment Complexity: Vertically integrating physical hardware constraints with formal kinematic logic significantly increases deployment friction compared to containerized, hardware-agnostic cloud architectures.
- Schema-Evolution Rigidity: Enforcing semantic firewalls via strict Protocol Buffers creates extreme rigidity; adding new sensing modalities requires recompiling the execution contracts across all 8 layers simultaneously.
- Federated Synchronization Overhead: Profile A micro-edge nodes operating in intermittent network states face unbounded convergence delays during global federated synchronization cycles.
- Constrained Model Flexibility: The irreversible dimensionality reduction at Layer 3 strictly prohibits the late-stage addition of new analytical features requiring historical raw data.

- **Hardware-Dependence Constraints:** The architecture assumes specific hardware execution characteristics (e.g., UMA for zero-copy memory transfers); performance and isolation guarantees degrade unpredictably on mismatched heterogeneous infrastructure.

PAPER-TO-LAYER TRACEABILITY

Table \ref{tab:trace} maps core papers from the ScholarMaster series to the architectural layers they enforce. Note: System-wide integration (P10), Deployment (P11/P12), and Verification (P18) papers span multiple boundaries.

TABLE: System Traceability Matrix

CONCLUSION

This unified reference architecture shows that achieving the combination of both privacy legitimacy and operational viability is not possible via ad-hoc subsystem optimization (e. g. , regulatory legal document alignment or even cloud-based access control). Instead, trust has to be composited into the silicon of the edge via Constraint-First Architectural Synthesis. Only by strict vertical integration of physical constraints, semantic firewalls, and sociotechnical visibility can we arrive at a coherent, unified modeling framework for the realisation of bounded

operational integration. In the context of the ScholarMaster, this unified reference architecture is the only architecture that implements a constraint-first approach while also providing an integrated systems synthesis subsystem, without adding new theorem proving, runtime crash handling, or machine learning optimisation components.

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